

The fruit fly *Drosophila melanogaster* is frequently used as a model system to investigate neurological disorders. As in humans, male flies have a single X chromosome and female flies have two X chromosomes.

The three genes *Abl*, *Enabled* (*Ena*), and receptor protein tyrosine phosphatase (*PTP*) are involved in axonal guidance during neuronal development and are all believed to operate in the same pathway. To investigate interactions between *PTP* and either *Ena* or *Abl*, fly lines with dominant mutant versions of each gene were developed and identified by eye characteristics. A dominant mutation that inactivates *PTP* (*P*) results in red and white eyes; a dominant *Ena* mutation (*E*) yields orange eyes (or red and orange if the *PTP* and *Ena* mutations are both present); and a dominant *Abl* mutation (*A*) yields elongated eyes. Using these fly lines, the following experiments were conducted (the recessive wild-type forms of each gene are indicated by lowercase letters).

Experiment 1

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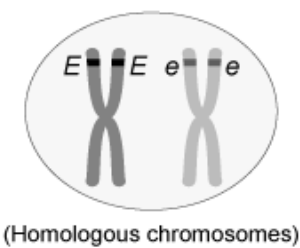
Experiment 2

PTP and *Abl* are located on the same chromosome. Females with the *PTP* mutation on one copy of the chromosome and the *Abl* mutation on the other copy were crossed with wild-type males. In *Drosophila*, recombination occurs only in female gametes. F1 Progeny with parental and recombinant genotypes are shown in Table 1.

Table 1 Phenotypes of 500 F1 Flies

Genotype	Eye phenotype	Number of flies
pApA (recombinant)	White, normal-sized eyes	22
pApA (parental)	White, elongated eyes	228
PApA (parental)	Red and white, normal-sized eyes	229
PApA (recombinant)	Red and white, elongated eyes	21

F1 males with red and white elongated eyes were then crossed with homozygous *PTP* mutant females expressing normal eye shape. This produced an F2 generation in which 50% of both male and female progeny had elongated eyes.



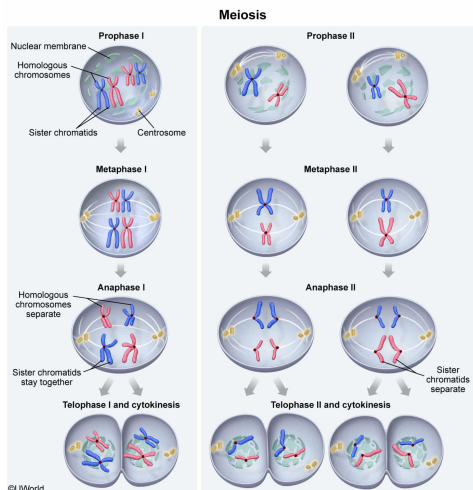
The diagram above represents the immature gamete of a male fly that is heterozygous for *Ena*. If the gamete undergoes meiotic division and *Ena* alleles do not undergo recombination, during which phase of meiosis will the sister chromatids containing the *e* allele be pulled to opposite poles of the cell?

- ☐ A. Metaphase I [2%]
- ☐ B. Anaphase I [18%]
- ☐ C. Metaphase II [5%]
- ☒ D. Anaphase II [72%]

Correct

72% answered correctly

Explanation:



The **formation of gametes** (ie, eggs, sperm) requires cellular division via **meiosis** rather than **mitosis**. Meiosis consists of two major stages, meiosis I and meiosis II. Like mitosis, each stage of meiosis is divided into prophase, metaphase, anaphase, and telophase. The homologous chromosomes first align at the metaphase plate in metaphase I and then move to opposite poles of the cell during anaphase I. Unlike anaphase of mitosis, which separates sister chromatids, **anaphase I of meiosis** separates homologous chromosomes.

Meiosis II then proceeds similarly to mitosis but **differs** in that each cell begins with a haploid, rather than a diploid, genome. During anaphase II of meiosis, **sister chromatids** are separated to opposite poles of the cell. Assuming no recombination, the identical *e* alleles on one set of sister chromatids will be pulled to opposite poles by the mitotic spindle during **anaphase II**.

(Choices A and C) Metaphase is the phase in which chromosomes align in the middle of the cell on the metaphase plate. Chromosomes, and therefore the alleles on them, are not pulled to the poles of the cell during this phase of mitosis or meiosis.

(Choice B) Because sister chromatids are copies of each other, they have identical *Ena* alleles. Therefore, in **anaphase I** in an *Ena* heterozygote, wild-type *Ena* alleles remain together on one set of sister chromatids, as do mutant alleles on the other set.

Educational objective:

Gamete formation requires cellular division by meiosis I and meiosis II, in which homologous chromosomes and sister chromatids, respectively, are separated. In the absence of recombination, maternal and paternal alleles are separated from each other during anaphase I, and identical alleles on each sister chromatid are separated from each other during anaphase II.

Time spent: 0 sec

Subject:
Biology

QID: 401615

System:
Genetics and Evolution

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Experiment 2

PTP and *Abl* are located on the same chromosome. Females with the *PTP* mutation on one copy of the chromosome and the *Abl* mutation on the other copy were crossed with wild-type males. In *Drosophila*, recombination occurs only in female gametes. F1 Progeny with parental and recombinant genotypes are shown in Table 1.

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In Experiment 1, approximately what fraction of the F2 generation is expected to die during the larval stage?

- ☐ A. 1/8 [10%]
- ☒ B. 3/8 [54%]
- ☐ C. 1/4 [16%]
- ☐ D. 9/16 [18%]

Correct

54% answered correctly

Explanation:

♀ EePp × ♂ EePp

Maternal contribution

	EP	Ep	eP	ep
EP	EEPP	EEpP	EePP	EePp
Ep	EEpP	EEpp	EePp	Eepp
eP	EePP	EePp	eePP	eePp
ep	EePp	Eepp	eePp	eepp

Paternal contribution

6/16 = 3/8

©UWorld = dies in larval stage

In any cross, each parent contributes one of two alleles per gene to each offspring. **Punnett squares** are used to predict the **distribution of genes** provided by two individuals with known genotypes. A Punnett square is set up by writing each possible combination of maternal alleles along the top of a grid and each possible combination of paternal alleles down the left side (or vice versa). The squares of the grid are then filled in with the corresponding maternal and paternal contributions, and each square represents an equally probable outcome.

The F2 generation in Experiment 1 was produced by a cross between F1 flies, each with an EePp genotype. According to the passage, either of the following is sufficient to cause death in the larval stage:

- Two mutant *Ena* alleles (ie, EE genotype)
- Two mutant *PTP* alleles *plus* at least one mutant *Ena* allele (ie, EePP genotype or EEPP genotype)

Of the 16 boxes in the Punnett square, 6 represent results that meet these criteria, meaning that 6/16, or 3/8, of the F2 generation are expected to die in the larval stage.

(Choice A) 1/8 is the expected number of progeny with 2 mutant *PTP* alleles and 1 mutant *Ena* allele, but this does not account for the offspring with 2 mutant *Ena* alleles.

(Choice C) 1/4 is the expected number of progeny with 2 mutant *Ena* alleles, but this does not account for the offspring with 2 mutant *PTP* alleles and 1 mutant *Ena* allele.

(Choice D) 9/16 is the number of F2 progeny expected to have at least one dominant *Ena* and one dominant *PTP* allele, but some of those offspring are heterozygous for *both* traits (EePp) and will survive to adulthood, just like their EePp parents from the F1 generation.

Educational objective:

Punnett squares show the expected distribution of genes passed on by parents of known genotypes. They are used to predict the percentage of an offspring generation that will have a given trait.

Subject.
Biology

System.
Genetics and Evolution

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Experiment 2

PTP and *Abl* are located on the same chromosome. Females with the *PTP* mutation on one copy of the chromosome and the *Abl* mutation on the other copy were crossed with wild-type males. In *Drosophila*, recombination occurs only in female gametes. F1 Progeny with parental and recombinant genotypes are shown in Table 1.

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Of the genes tested in the passage, which are autosomal?

- I. *Ena*
- II. *Abl*
- III. *PTP*

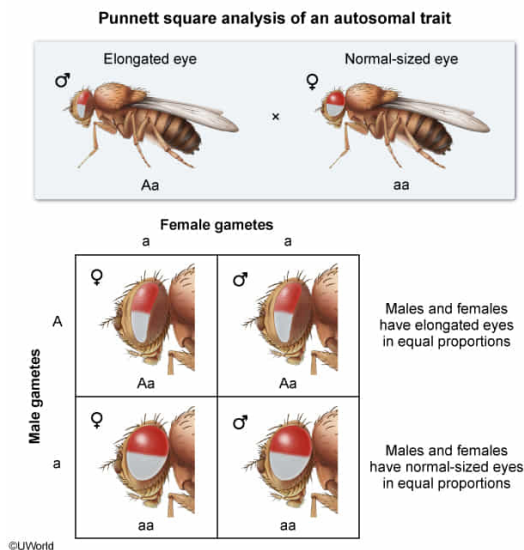
- ☐ A. II only [7%]
- ☐ B. I and III only [10%]
- ☐ C. II and III only [21%]

✔ ☒ D. I, II, and III [59%]

Correct

59% answered correctly

Explanation:



In mammals and many insects, genetic traits are generally passed to the next generation in either an **autosomal** or a **sex-linked** manner. Of the chromosomes in a zygote, two are sex chromosomes (X,Y) and the remaining ones are called autosomes.

- Sex-linked traits arise due to the expression of alleles present on an organism's sex chromosomes. Because the X chromosome contains significantly more genes than the Y chromosome, the majority of sex-linked traits are linked to the X chromosome. Due to **sex determination**, male offspring (XY) receive their X chromosome from their mother (XX) and their Y chromosome from their father (XY). Because males have only one X chromosome and cannot compensate with another, males in a population generally express **recessive X-linked** traits in **higher proportions** than their female counterparts.
- Autosomal traits arise due to the expression of alleles present on an organism's autosomes. These alleles are passed to offspring through inheritance of the parental autosomes. Because males and females all have the same number of autosomes, autosomal traits are generally expressed in the **same proportions** in males as in females.

The passage states that males in the first cross of Experiment 1 were heterozygous for *Ena*, suggesting that they had two different copies of the gene. Therefore, *Ena* cannot be on the X chromosome and must be autosomal (**Number I**).

In Experiment 2, F2 males and females have the same ratios of normal-shaped eyes to

elongated eyes. Based on the cross used to generate the F2 generation, **if *Abi* were sex-linked** all females would have elongated eyes (dominant) and all males would have normal eyes (recessive). Because this is not the case, *Abi* must be autosomal (**Number II**). Because *PTP* is on the same chromosome as *Abi*, it must also be autosomal (**Number III**).

Educational objective:

Genetic traits may be autosomal or sex-linked (X chromosome). Because males have only one

X chromosome, sex-linked recessive traits tend to be present at greater rates in males than in females, whereas autosomal traits are present in both at the same proportions.

Time spent:0 sec

Subject:
Biology

QID:401617

System:
Genetics and Evolution

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Experiment 2

PTP and *Abl* are located on the same chromosome. Females with the *PTP* mutation on one copy of the chromosome and the *Abl* mutation on the other copy were crossed with wild-type males. In *Drosophila*, recombination occurs only in female gametes. F1 Progeny with parental and recombinant genotypes are shown in Table 1.

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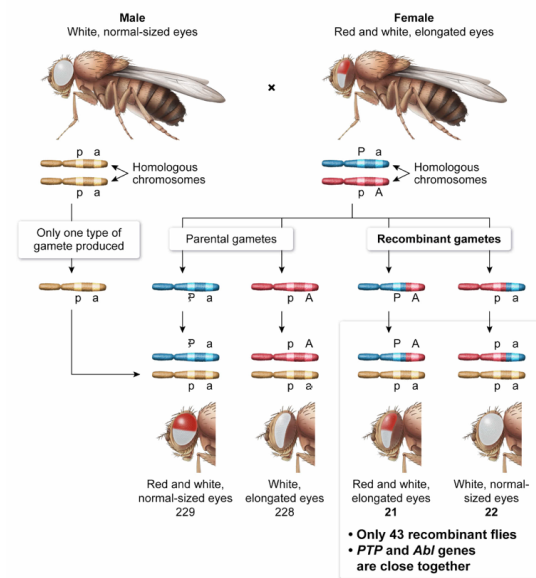
Which evidence in Table 1 suggests that *PTP* and *Abl* are located close together on the chromosome?

- ☐ A. The two recombinant genotypes occur with similar frequency. [38%]
- ☒ B. Only a small percentage of the progeny are recombinant. [54%]
- ☐ C. Recombination did not occur in male gametes. [2%]
- ☐ D. The F1 generation has a high recombination frequency. [4%]

Correct

53% answered correctly

Explanation:



Homologous chromosomes can exchange genetic information in a process called **crossing over**, or **recombination**, during prophase I of meiosis. Crossing over can produce new combinations of alleles within a chromosome by moving some alleles of the maternal copy to the paternal copy and vice versa. New combinations of alleles are called *recombinant*, whereas combinations that already existed in the parent are called *parental*.

If two genes are **located close together** on a chromosome (measured in centimorgans, or **map units**), they are relatively **unlikely to be separated** by a recombination event. This is because there is less distance between the two genes in which a recombination can occur. Therefore, substantially fewer progeny will have recombinant combinations than parental combinations. As a result, the **recombinant progeny** can be identified by the fact that there are **fewer of them**. Table 1 shows that only 43 of 500 flies (~9%) were recombinant. Given this small percentage of recombinant flies, it is likely that the *PTP* and *AbI* genes are **close together on the chromosome**.

(Choice A) Recombination is the exchange of genes between homologous chromosomes and always produces two recombinant chromosomes. Because each chromosome is equally likely to be passed on, recombinant genotypes *a/ways* occur with equal frequency, regardless of genetic distance.

(Choice C) According to the passage, recombination occurs only in female gametes. However, the lack of recombination in male gametes does not provide any information about the distance between genes on a chromosome.

(Choice D) A high recombination frequency would indicate that genes are *distant* from, not close to, each other on the chromosome.

Educational objective:

Homologous chromosomes can exchange genetic material by recombining. Genes that are located close together on a chromosome have a lower probability of being separated by

recombination than those that are far apart. Fewer progeny from a cross will have recombinant genotypes than will have parental genotypes.

Time spent:0 sec

Subject:
Biology

QID:401618

System:
Genetics and Evolution

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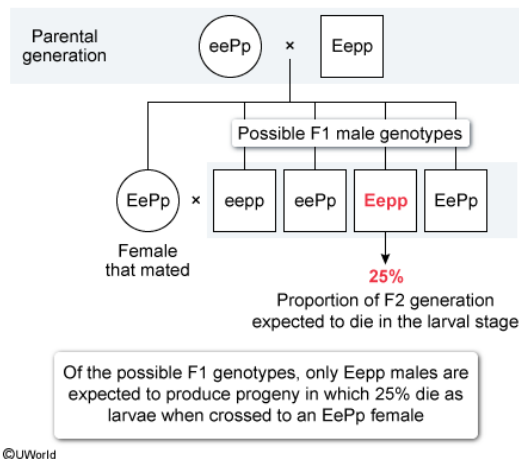
Suppose a female heterozygous for both *PTP* and *Ena* (*EePp*) was unintentionally mated with a single F1 male of an unknown genotype from Experiment 1. If 25% of the resulting F2 progeny died in the larval stage, the genotype of the F1 male was most likely:

- ✔ ☒ A. *Eepp* [39%]
- ☐ B. *eePP* [26%]
- ☐ C. *eepp* [8%]
- ☐ D. *EePP* [25%]

Correct

40% answered correctly

Explanation:



Pedigree charts can be used to determine the **possible outcomes of various matings**. When two individuals of known genotypes mate, the possible offspring genotypes can be determined in the pedigree based on the alleles available. Similarly, if the genotype distribution of the offspring is known, the parental genotypes can often be determined.

According to the passage, flies with $EEPP$, $EEpp$, $EePP$, or $EePp$ genotypes die as larvae. The F1 generation in Experiment 1, from which the $EePp$ female was selected, was the product of a cross between an $eePp$ female and an $Eepp$ male. Based on these parental genotypes, the F1 generation has only **four possible genotypes** (including the $EePp$ genotype that was *intended* to be used): $EePp$, $Eepp$, $eePp$, and $eepp$. Consequently, two genotypes given in the answer choices can be ruled out because they are not present in the F1 generation:

- **Choice B can be ruled out** because, although 25% of progeny *would* die in this cross if it occurred, flies with the genotype $eePP$ *cannot* be present in the given F1 generation. According to the passage, the father of the F1 generation was $Eepp$, and could contribute only the recessive p allele. Therefore, all F1 flies must have at least one recessive p allele.
- **Choice D can be ruled out** for the same reason: $EePP$ flies are absent from F1. In addition, $EePP$ flies would not survive to adulthood (ie, mating age) even if they *were* present in F1.

Choice C can be ruled out because if the $EePp$ F1 female had **mated with an $eepp$ F1 male**, *all progeny* would be expected to survive to adulthood (ie, each would receive at least one e allele and one p allele).

The only other answer choice is $Eepp$. A **Punnett square** confirms that 25% of the progeny from a cross between an **$Eepp$** male and the $EePp$ female are expected to die in the larval stage because they would inherit $EEpp$ or $EEpp$ genotypes.

Educational objective:

Pedigree charts can help determine either the possible genotypes of the progeny if the parental genotypes are known or the possible parental genotypes if the genotypes of the progeny are known.

Time spent:0 sec

Subject:

Biology

QID:401619

System:

Genetics and Evolution

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Axonal guidance is the process through which axons reach their proper targets in the brain. Researchers hypothesize that the wild-type *PTP* gene product mediates proper axonal guidance by inactivating the *Abl* gene product. Which experimental

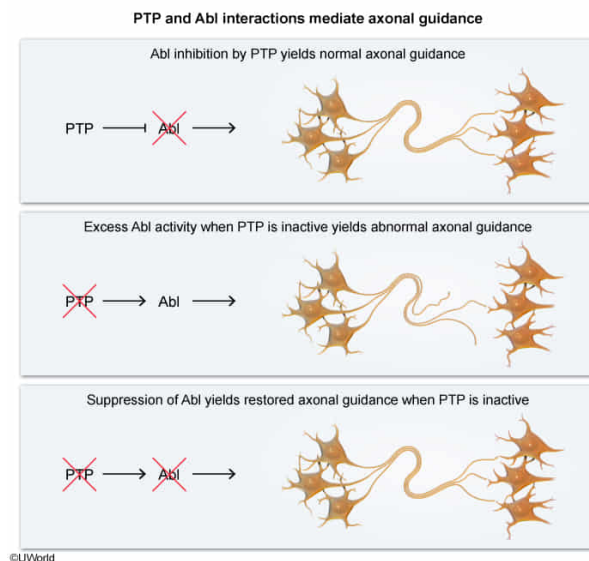
observation would best support this hypothesis?

- ☐ A. Suppression of *Abl* increases the severity of abnormal axonal guidance in *PTP* mutants. [9%]
- ☐ B. Overexpression of *Abl* reduces the severity of abnormal axonal guidance in *PTP* mutants. [8%]
- ☒ C. Suppression of *Abl* reduces the severity of abnormal axonal guidance in *PTP* mutants. [79%]
- ☐ D. Overexpression of *Abl* restores normal axonal guidance in *PTP* mutants. [2%]

Correct

78% answered correctly

Explanation:



When two gene products function in the same biological pathway, altering an **upstream product** will have an **effect** on the activity of **downstream** targets or products. Altering a downstream product in a way that **mimics the function** of the upstream product can **compensate for the loss** of an upstream product.

Researchers hypothesized that the function of *PTP* in axonal guidance is to inactivate *Abl*. If this is correct, then when *PTP* is inactivated by a mutation, *Abl* will be overly active, leading to abnormal axonal guidance. Therefore, *Abl* must be suppressed to some extent for normal axonal guidance to occur, and its suppression in the presence of inactive *PTP* would help restore normal axonal guidance. In effect, suppression of *Abl* activity carries out the same function that active *PTP* does and therefore should lead to the same outcome.

(Choice A) Based on the researchers' hypothesis, abnormal axonal guidance occurs when *PTP* fails to suppress *Abl*. Therefore, *Abl* suppression normally helps maintain correct axonal guidance and would be unlikely to increase the severity of the

phenotype.

(Choices B and D) If abnormal axonal guidance is a result of too much *Abi* activity, **overexpressing *Abi*** in a background that already has too much *Abi* activity (ie, inactive *PTP*) would only increase the severity of the abnormal phenotype. It would not restore the normal state.

Educational objective:

Upstream gene products in a biological pathway act on downstream gene products. Failure of an upstream product to act can be compensated for by alterations to downstream targets that mimic the effect of the upstream product.

Time spent:0 sec

Subject:

Biology

QID:401620

System:

Genetics and Evolution